Juan Felipe Ariza Mejía

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Education

Ph.D. in Mathematics, The University of Iowa, Iowa City, USA 2021 - July 2027 (expected)
Advisor: Prof. Ionuţ Chifan

MMATH in Pure Mathematics, University of Waterloo, Waterloo, Canada 2019 - 2020
Advisor: Prof. Nico Spronk

Bs.C. in Mathematics, Universidad de los Andes, Bogotá, Colombia 2013 - 2018
Advisor: Prof. Stefano Ferri

Bs.C. in Physics, Universidad de los Andes, Bogotá, Colombia 2013 - 2018
Advisor: Prof. Andrés F. Reyes Lega (co-advisor: Dr. Mitsuru Wilson)

Research Interests

Operator Algebras – Structural properties for von Neumann algebras - Rigidity for group von Neumann algebras and group C*-algebras

Group Theory – Analytic and approximation properties for groups - Geometric Group Theory

Ergodic Theory – Orbit equivalence relations - Measured Group Theory

Publications – Preprints

1. *Amenable absorption in von Neumann algebras of hyperbolic groups*, J. F. Ariza Mejía, I. Chifan A. Fernández Quero and A. Ioana. arXiv:2606.10105
2. *Non-amenable C^*-superrigid groups that are not W^*-superrigid*, J. F. Ariza Mejía, I. Chifan and A. Fernández Quero. arXiv:2602.05290
3. *McDuff superrigidity for group II_1 factors*, J. F. Ariza Mejía, I. Chifan, D. Osin and B. Sun. arXiv:2511.23123
4. *Relative solidity results and their applications to computations of some* II_1 *factor invariants*, D. N. Amaraweera Kalutotage, J.F. Ariza Mejía, I. Chifan and K. Khan. arXiv:2509.19481

Selected Talks

1. Contributed talk at Great Plains Operator Theory Symposium, The University of Iowa, May 2026
2. JMM 2026, AMS Special Session on Advances in Operator Algebras I, Washington D.C., January 2026
3. University of California, San Diego – Functional Analysis Seminar, December 2025
4. The University of Iowa – Operator Theory Seminar, September 2025
5. AMS 2025 Spring Central Sectional Meeting, Special Session on Advances in Operator Algebras I, University of Kansas, March 2025
6. Invited lightning talk at Group Operator Algebras: Classification, Structure and Rigidity, Banff International Research Station, Canada, September 2024
7. Invited lightning talk at YMC*A: Young Mathematicians in C^*-algebras, University of Glasgow, August 2024

8. Two-part invited expository talk at Groundwork for Operator Algebras Lecture Series, IPAM Institute for Pure and Applied Mathematics, Los Angeles, July 2024
9. Contributed talk at Great Plains Operator Theory Symposium, University of Nebraska-Lincoln, June 2024
10. Purdue University – Operator Algebra Seminar, February 2024
11. The University of Iowa – Operator Theory Seminar, November 2023

Teaching

Instructor on Record

Department of Mathematics, The University of Iowa

College Algebra	Fall 2026
Calculus III	Spring 2026
Elementary Functions	Fall 2023
Analysis Qualifying Exam Preparation Seminar (co-instructor)	Summer 2023
Trigonometry	Spring 2023
College Algebra	Fall 2022

Teaching Assistant

Department of Mathematics, The University of Iowa

Foundations of Analysis	Fall 2025
Engineer Math I: Single Variable Calculus	Spring 2025
Introduction to Analysis II (helper) – *1st year graduate course for which I held weekly office hours and helped with weekly assignments solving doubts and giving feedback*	Spring 2025
Calculus I	Fall 2024
Introduction to Analysis I (helper) – *1st year graduate course for which I held weekly office hours and helped with weekly assignments solving doubts and giving feedback*	Fall 2024
Engineering Math II: Multivariable Calculus	Spring 2022
Engineer Math II: Multivariable Calculus	Fall 2021
Calculus for Biological Sciences	Fall 2021

Department of Pure Mathematics, University of Waterloo

Calculus I for Sciences	Spring 2020
Calculus II for Engineering	Spring 2020
Introduction to Real Analysis	Winter 2020
Calculus II (Advanced)	Winter 2020
Linear Algebra	Fall 2019

Department of Mathematics, Universidad de los Andes

Integral Calculus with Differential Equations	First Semester 2017
Linear Algebra	Second semester 2017

Department of Physics, Universidad de los Andes

Physics Problem Solving Center helper	First semester 2016

Other Conferences and Workshops attended

Virginia Operator Algebras Workshop, University of Virginia — July 2026

ECOAS 2025 - East Coast Operator Algebras Symposium, University of Nebraska - Lincoln — October 2025

Workshop on Operator Algebras and Applications: Groups and Group Actions, Fields Institute, Toronto — October 2023

YMC*A 2023 - Young Mathematicians in C*-algebras, KU Leuven — August 2023

NCGOA Spring Institute 2023 - Non-commutative Geometry and Operator Algebras, Vanderbilt University — May 2023

Modern Trends in Operator Algebras Conference, UCLA — March 2023

Workshop on Von Neumann Algebras and Geometric Group Theory, UCSD — February 2023

ECOAS 2022 - East Coast Operator Algebras Symposium, Michigan State University — October 2022

YMC*A 2022 - Young Mathematicians in C*-algebras, University of Oslo — August 2022

GOALS 2022 - Groundwork for Operator Algebras Lecture Series, University of Montana — July 2022

COSy 2022, Canadian Operator Symposium, University of Ottawa — June 2022

Topological Quantum Groups, C*-Tensor Categories, and Subfactors Summer School and Workshop, University of Waterloo — May 2022

Service

Co-webmaster, Great Plains Operator Theory Symposium, The University of Iowa — May 2026

Co-organizer and co-webmaster, UIowa Operator Algebras Research Seminar — Fall 2025

Contributed lecture at GOALS 2025 - Groundwork for Operator Algebras Lecture Series (online)
W*-lecture 2.1 Abelian von Neumann Algebras — June 2025

Invited Expository Speaker at GOALS 2024, IPAM, Los Angeles — July 2024

Graduate Student Mentor for the Directed Reading Program — Fall 2022, Fall 2023, Fall 2024

Graduate Student Senate (Mathematics alternate senator) — August 2024 - August 2025

Graduate Student Senate (Mathematics senator) — August 2023 - August 2024

Co-organizer Jakobsen Graduate Research Showcase — March 2024

Co-organizer Operator Theory and Operator Algebras Student Seminar — Fall 2023, Fall 2025

Co-creator Operator Theory and Operator Algebras Student Seminar — Spring 2023

Selected Honors and Awards

Ballard and Seashore Dissertation Fellowship — Spring 2027
Graduate College, The University of Iowa

Post-Comprehensive Research Fellowship — Spring 2024
Graduate College, The University of Iowa

Erwin and Peggy Kleinfeld Scholar — Fall 2023 - Spring 2026
Mathematics Department, The University of Iowa

Kleinfeld Research Scholarship (for passing two Ph.D. qualifying exams upon admission) — Fall 2021
Mathematics Department, The University of Iowa

Kleinfeld Math Graduate Recruitment Fellowship — Fall 2021, Summer 2022, Summer 2023
Mathematics Department, The University of Iowa

University of Waterloo Graduate Scholarship — Spring 2020
University of Waterloo

Mathematics Faculty Graduate Award — Winter 2020
University of Waterloo

Joseph Wai-Hung Liu Graduate Scholarship — Winter 2020
University of Waterloo

University of Waterloo Graduate Scholarship — Fall 2019
University of Waterloo

Alberto Magno award, best undergraduate admission score — Second semester 2013
Universidad de los Andes